Violin Plots

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Introduction

A violin plot displays a kernel-density estimate vertically and reflects it around the central axis, producing the symmetric outline that gives the geom its name. It combines the clarity of a density plot with the side-by-side grouping that boxplots provide, and — crucially — reveals multi-modality and skew that boxplots flatten into a single rectangle. For comparing the *shape* of distributions across categories, violin plots are often the right default; for comparing only the median and quartile structure, boxplots remain more compact.

Prerequisites

A working understanding of kernel density estimation, boxplots, and `ggplot2` aesthetic mappings.

Theory

For each group, `geom_violin()` computes a kernel density $\hat{f}(x)$ on the response variable, draws the density curve vertically, and mirrors it across the group's central axis. The resulting width at each y-value is proportional to $\hat{f}(y)$, so wide regions correspond to high density. The `scale` argument controls how groups are sized relative to each other: `"area"` makes all violins equal area, `"count"` scales width proportional to the group's sample size, and `"width"` makes them all the same maximum width. The `trim` argument decides whether the density tails are truncated at the data extremes (`TRUE`, default) or extrapolated beyond them (`FALSE`).

Assumptions

A continuous outcome and a categorical (or ordered) grouping variable, with enough observations per group (at least 20–30) for the kernel density to be meaningful. Below that threshold the shape becomes unstable and a strip plot or jittered scatter is more honest.

R Implementation

```r
library(ggplot2)

# Basic violin
ggplot(iris, aes(Species, Sepal.Length, fill = Species)) +
  geom_violin() +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()

# Violin + boxplot overlay
ggplot(iris, aes(Species, Sepal.Length, fill = Species)) +
  geom_violin(alpha = 0.5, trim = FALSE) +
  geom_boxplot(width = 0.2, fill = "white", outlier.shape = NA) +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()
```

```r
# Violin + jittered points
ggplot(iris, aes(Species, Sepal.Length, fill = Species)) +
  geom_violin(alpha = 0.5) +
  geom_jitter(width = 0.1, alpha = 0.5) +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()
```

Output & Results

Three views of the iris sepal-length data: a plain violin per species, a violin with a narrow boxplot overlay showing median and quartiles, and a violin with jittered raw points overlaid for full transparency. The third version is the most informative when the audience is unfamiliar with kernel-density visualisations.

Interpretation

A reporting sentence (figure caption): "Sepal length by species (n = 50 per group); violin width is proportional to kernel-density estimate (`scale = "area"`), the inset boxplot summarises the median and IQR, and jittered points show individual observations." Always describe what the violin's width encodes; readers unfamiliar with the geom default to thinking it represents count.

Practical Tips

- For multi-modal distributions (e.g. Old Faithful eruption times), the violin reveals the second mode that a boxplot would hide entirely; this is the strongest single argument for the geom.
- Set `trim = FALSE` only when the density tails are biologically meaningful; otherwise the default trim avoids implausible extrapolation.
- Combine with a narrow boxplot (`width = 0.15`) when readers expect numeric summaries; the inset preserves quartile information without dominating the visual.
- For paired or matched groups, `introdataviz::geom_split_violin()` produces left-right halves so two conditions share the same x-position.
- For large n, prefer violins over jittered points; violins scale to millions of observations whereas individual points stack into uninformative blobs.
- For small n (< 20 per group), a strip plot or a boxplot is more honest than a violin; the kernel density estimate is too noisy to trust.

R Packages Used

`ggplot2` for `geom_violin()`; `introdataviz` for split violins; `ggdist` for `stat_halfeye()` and other slabinterval geoms that combine violins with credible-interval annotations; `vioplot` for a base-R alternative when `ggplot2` is not available.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- ggplot2 grammar and multi-panel layouts